

ANM DDR5 Module Family

Synthetic technical datasheet for Anistroph and Aina-Veris testing. Product identifiers are fictional; engineering parameters use realistic memory-industry domains.

Product Summary

Product	Tech	Density	Rate	Width	Temp	Package
ANM-RD5-0001	DDR5	32 GB	6400 MT/s	x80	-40..95 C	TFBGA
ANM-RD5-0002	DDR5	128 GB	6400 MT/s	x72	0..95 C	VFBGA
ANM-RD5-0003	DDR5	16 GB	5600 MT/s	x80	-40..95 C	VFBGA
ANM-RD5-0004	DDR5	128 GB	6400 MT/s	x80	-40..95 C	VFBGA
ANM-UD5-0001	DDR5	32 GB	5600 MT/s	x64	0..95 C	VFBGA
ANM-UD5-0002	DDR5	16 GB	6400 MT/s	x64	0..95 C	VFBGA
ANM-UD5-0003	DDR5	64 GB	6400 MT/s	x64	-40..95 C	VFBGA
ANM-UD5-0004	DDR5	64 GB	5600 MT/s	x64	0..95 C	VFBGA

Power Management

DDR5-class products support controller-managed idle, power-down, and self-refresh states. Entry and exit timing must match the configured operating point. Higher transfer rates increase the importance of power-delivery integrity and thermal margin. LPDDR5X products use programmable low-power behavior with additional controller-managed state transitions.

Initialization

After stable power and clock conditions, the controller performs reset release, mode-register configuration, timing setup, and required training before normal traffic. The timing profile must correspond to the selected transfer rate; lower-speed timing assumptions must not be reused at higher operating points.

Temperature and Operating Restrictions

A required operating point is valid when it falls inside the product's stated minimum and maximum temperature range. For example, a 55 C requirement is satisfied by a product rated 0..95 C or -40..95 C. The value 55 C is therefore interpreted as a range-containment requirement rather than a discrete catalog grade.

Family Selection Notes

Products in this family intentionally vary by density, data rate, organization, temperature range, and package. Hard engineering constraints should be applied through structured parametric search; this document is intended for qualitative technical comparison after shortlisting.

Revision S1 - synthetic reference document; not a commercial component datasheet.